# Band-1 Hadamard Order 10112 – Classical Paley-I (q=10111); IN SPK via Paley-I; Gap 2052 Still OPEN

H_10112 Paley-I PASS; Paley-II FAIL/SKIP (q=5055=3*5*337 NOT prime / not prime power); Kronecker FAIL/SKIP (4*2528; 2528===0 mod 4 admissible BUT no on-box H_2528); Miyamoto digression skipped; gap 2052 OPEN; conjecture OPEN; GATE-04 LOCKED

C.M. Baird; ZoraASI mission-control Band-1 dash

2026-09-23

**Document class.** Scientific computational / construction certificate (Band-1).
**Program.** Open-mathematics collapse list * Human+AI attack order * ZoraASI mission-control.
**Parents.** `2026-09-23_Band1_Hadamard_10104_PaleyI`; `2026-09-23_Band1_Hadamard_10100_PaleyI`; `2026-09-23_Band1_Hadamard_10068_PaleyI`; `2026-09-23_Band1_Hadamard_10040_PaleyI`.
**Lab.** `/workspace/experiments/band1/hadamard_10112_paleyI/`.
**Cutoff.** 2026-09-23T21:00:00Z (~5:00PM ET).

**Honesty stamp (non-negotiable).**
This memo claims **finite construction of order 10112 only (verified max|HHˆT-nI|=0), via classical Paley-I q=10111 (prime ===3 mod 4). 10112 IN SPK via Paley-I.** Honest skip: Miyamoto digression this wave (prefer classical SPK). Classical SPK for 10112: Paley-I **PASS** ($q = 10111$ prime $\equiv 3 \pmod 4$); Paley-II **FAIL/SKIP** ($q = 5055 = 3 \cdot 5 \cdot 337$ NOT prime / not prime power); Kronecker **FAIL/SKIP** (note $10112 = 4 \cdot 2528$; $2528 \equiv 0 \pmod 4$ admissible BUT no on-box H_2528). **10112** folds into SPK **only** via classical Paley-I (this run). It **does not** claim the full Hadamard conjecture. GATE-04 collection remains **LOCKED**. No RH / P≠NP / YM / Frankl / NS-solved / AC-solved claims. Gap **2052** remains **OPEN**. `clay_closed = false`. Prior SPK **10104+10100+10068+10040+10008+9968+9932+9924+9908+9888+9884+9872+9860+9852+9840+** retained; Miyamoto-only **8948/8644/8516/8452/7988/...** excluded.

ewpage

## Abstract

Finite construction of order **10112**.

- $H_{10112}$ = Paley-I ($q = 10111$) – classical SPK
- Paley-II for 10112: **FAIL/SKIP** ($q = 5055 = 3 \cdot 5 \cdot 337$ NOT prime / not prime power)

- Kronecker for 10112: **FAIL/SKIP** ($4 \cdot 2528$; $2528 \equiv 0 \pmod 4$ admissible BUT no on-box H_2528)
- Skip: Miyamoto digression this wave

**SPK: IN SPK via Paley-I**.
**Gap 2052:** OPEN. **Conjecture:** OPEN. **clay_closed:** false. **GATE-04:** LOCKED.

```
python3 verify_hadamard_10112.py
```

**Results.** overall_pass=True; units 202/202; row blocks 79/79 (block size 128); wall 848.1475s; constructed max|HHˆT-nI|=0.

Cutoff: 2026-09-23T21:00:00Z (~5:00PM ET)

## Arithmetic

- Target $n = 10112 = q + 1$ with $q = 10111$ prime $\equiv 3 \pmod 4$ — Paley-I eligible.
- Paley-II: $q = 5055 = 3 \cdot 5 \cdot 337$ NOT prime / not prime power — FAIL/SKIP.
- Kronecker: $10112 = 4 \cdot 2528$; $2528 \equiv 0 \pmod 4$ admissible BUT no on-box H_2528 — FAIL/SKIP.
- Miyamoto digression skipped this wave (prefer classical SPK).

## Constructions (verified)

| Object | Method | max\|HHˆT-nI\| | Wall (s) |
|---|---|---|---|
| H_10112 Paley-I | Paley-I q=10111 | 0 | 829.4528 |
| H_10112 Paley-II | FAIL/SKIP (q=5055=3*5*337 NOT prime / not prime power) | – | – |
| H_10112 Kronecker | FAIL/SKIP (4*2528; 2528===0 mod 4 admissible BUT no on-box H_2528) | – | – |

## SPK closure note

Seeds include prior SPK through **10104** lineage (**10104**; **10100**; **10068**; **10040**; **10008**; **9968**; **9932**; **9924**; **9908**; **9888**; **9884**; **9872**; **9860**; **9852**; **9840**; **9812**; **9804**; **9792**; **9788**; **9768**; **9744**; **9740**; **9720**; **9680**; **9644**; **9632**; **9624**; **9620**; **9588**; **9552**; **9548**;

**9540**; **9512**; **9492**; **9480**; **9468**; **9420**; **9404**; **9392**; **9372**; **9344**; **9324**; **9320**; **9312**; **9284**; **9240**; **9228**; **9204**; **9200**; **9188**; **9152**; **9104**; **9128**; **9124**; **9092**; **2220+8884**; **2236**; **2160**; **2140+8564**; ...; **NOT** 8948/8644/8516/...) + verified **10112** (+ prior **10104+10100+10068+10040+10008+9968+9932+9924+9908+9888+9884+9872+9860+9852+9840+**
Next constructive gap **2052** (OPEN).

| Quantity | Value |
|---|---|
| 10112 in SPK | **True** (Paley-I q=10111) |
| Gap 2052 | **OPEN** |
| clay_closed | false |
| GATE-04 | LOCKED |

## Honesty

- clay_closed = false
- GATE-04 LOCKED
- Hadamard conjecture OPEN
- Gap 2052 OPEN (method triage / non-finding only)
- No RH / P!=NP / YM / Frankl / NS / AC solved claims
- SPK membership only via classical Paley-I/II/Kronecker (not Miyamoto alone)

## Next optional

q=10139->n=10140 Paley-I DIRECT q=10139 OR q=10151->n=10152 Paley-I DIRECT q=10151 OR q=10159->n=10160 Paley-I DIRECT q=10159 OR n=2052 standard SPK constructive gap – method-attack / lit OR Track B / Object 1 meter await